



Washington University in St. Louis

2024-25 AIAA Design/Build/Fly Proposal



1 Executive Summary

This proposal outlines Washington University in St. Louis' (WashU) preliminary design, manufacturing, and testing plan for "Ursa Major" to compete in the 2024/2025 AIAA Design/Build/Fly Competition. WashU Design/Build/Fly (WUDBF) is organized into six sub-teams to facilitate the successful completion of the objective to design, build, and fly an electric remote-control aircraft simulating an X-1 Supersonic Test Flight program through four different missions.

WUDBF's strategy is to maximize the weight of the payload and wing area of the aircraft, while minimizing the weight of the X-1 test vehicle in order to achieve the higher bonus-box score. In service of this objective the chosen design is a high wing configuration with a rectangular fuselage and single propeller propulsion system. The NACA 2412 airfoil will be paired with a Hershey Bar wing with a wingspan of 6 feet and a chord length of 1.457 feet. Due to the lack of takeoff distance constraint, a conventional tail with a bow-tricycle landing gear will be employed. Furthermore, to accommodate necessary electronic components and a potential internal fuel tank while minimizing weight, the fuselage will have a height of 6 inches, a width of 6 inches, and a length of 28 inches. WUDBF aims to carry 5.51 pounds of fuel for Mission 2 (M2) and use an X-1 test vehicle with a maximum weight of 0.25 pounds for Mission 3 (M3). The aircraft will place M2 fuel tank payloads underneath the wings for minimal effect on aerodynamic performance of the main aircraft. The M3 glider will have a maximum footprint of 1.5 feet wide, 1.5 feet long, and 3 inches tall. The propulsion pack will consist of a LiPo 8s (29.6-volt) 3250 milliamp hours battery, a SunnySky x3530 KV445 motor, and a 13x8 propellor.

WUDBF determines that it is best to minimize the weight of the glider in order to maximize M3 score. For M2, a balance between a higher fuel tank payload and wing area, with a lower cruise velocity will result in a maximized score. The team will use an iterative design process to transition from the prototype to the final aircraft.

2 Management Summary

2.1 Organizational Structure: WUDBF is a student run club with a faculty advisor. The executive board is split into both administrative and technical functions to ensure efficient workflow and operations. The roles and skills of each sub-team are further explained in Table 1. The organization of the sub-teams is depicted in Fig. 1.

Sub-Teams	Tasks	Skills
Aerodynamics	Design of the exterior geometry of the aircraft, as well as optimize aircraft stability.	Knowledge of aircraft performance, stability and control, CFD(Ansys), XFLR5.
Avionics	Installation of electrical components, programming of flight controllers, and test flight data collection.	Knowledge in electrical design, soldering, Real Flight 8, data collection.
Manufacturing	Assembly of the aircraft and components in addition to materials testing.	Knowledge in lasercutting, fiberglass layups, hot-wire foam cutting, 3D printing, and balsa aircraft construction.
Propulsion	Design of the propulsion package and motor testing.	Knowledge in propulsion theory and analysis, Ecalc, and RC Benchmark.
Simulation Development	Development of simulation tools used for optimization and sensitivity analysis.	Knowledge in MATLAB/Simulink and python.
Structures	Design and modeling of the aircraft structure and any additional mission specific components.	Knowledge in CAD, structural FEA, and mass properties analysis.

Table 1 Sub-team roles.

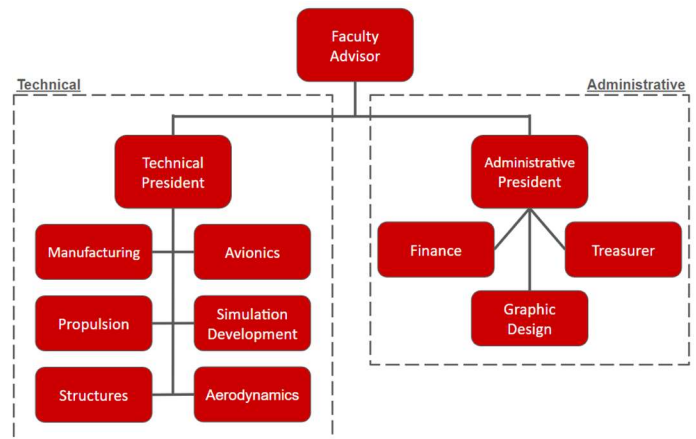


Figure 1 Sub-team structure.

Administrative responsibilities such as acquiring sponsorships, managing the budget, and graphic design for club marketing are overseen by the Administrative President. The Administrative President is also responsible for scheduling and leading general body meetings. The Administrative President manages administrative division, which includes three sub-teams: finance, graphic design, and treasury. The Technical President is responsible for approving technical decisions and oversees the technical division.

The technical division includes six sub-teams: simulation and development, aerodynamics, structures, propulsion, manufacturing, and avionics. Each sub-team is managed by a sub-team lead who organizes meeting tasks and ensures their sub-team is on track to complete their components of the overarching design and testing process in a timely manner.

2.2 Schedule: The team's proposed schedule is depicted below by the Gantt chart in Fig. 2. WUDBF adheres to the chart to meet deadlines for the design and testing process. The executive board and leadership team meet weekly to ensure effective cross-functional communication.

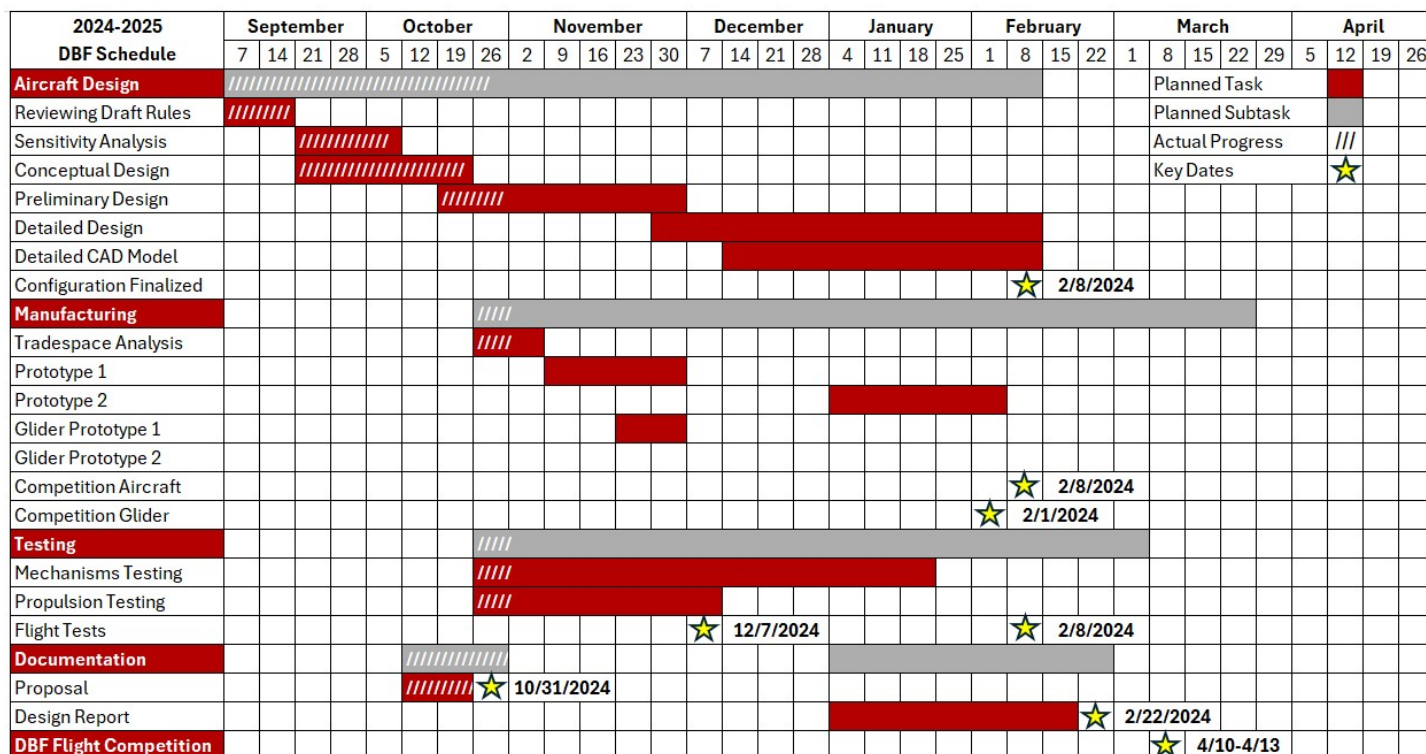


Figure 2 Proposed 2024-2025 plan.

2.3 Budget: WUDBF receives funding from two sources: Washington University Student Union (SU) and the Washington University McKelvey Grant. Expenses are based on previous years' spending as well as necessary line-itemed purchases. A summary of the allocated funds is shown in Table 2. The transportation cost includes the estimated cost of transporting the plane to Tucson, AZ. The SU materials budget includes structural construction materials such as wood, fiberglass, glues, and epoxies. The electronics budget encompasses all necessary batteries, motors, servos, and wires. The McKelvey Grant budget is used for materials and equipment to test and improve on the current aircraft construction techniques.

Description	Cost
Construction	\$ 9,423.61
Materials	\$ 5,628.64
Electronics	\$ 3,794.97
Travel	\$13,317.60
Flights	\$ 7,200.00
Rental Cars	\$ 3,237.60
Lodging	\$ 2,880.00
McKelvey Grant	\$ 4,900.00
Project Expenses	\$ 1,600.00
Materials	\$ 2,300.00
Transportation	\$ 1,000.00
Misc.	\$ 720.00
AIAA Memberships	\$ 720.00
Total	\$28,361.21

Table 2 2024-2025 Budget.

3 Conceptual Design

3.1 Mission Requirements: The competition has a X-1 Supersonic Flight Test Program theme, involving the launch of an autonomous glider. Competition is broken up into three flight missions and one

ground mission. The mission requirements are detailed in Table 3. The team has determined glider weight and bonus box score to be highest priority for a successful M3, while M2 relies more on a balance of heavy fuel weight and quick lap time. The team has decided to optimize for M3.

Mission	Scoring	Requirements	Sub-System Design Considerations
M1: Delivery Flight	1.0	Aircraft must be put in flight configuration within 5 minutes. Aircraft must then complete 3 laps in 5 minute flight window.	Fuel tanks and X-1 test vehicle must attach easily and securely to aircraft.
M2: Captive Carry Flight	$1 + N[\frac{\text{fuel weight}}{\text{time}}]$	Fuel tanks and flight battery must be installed in 5 minutes. Aircraft must fly 3 laps quickly with weight.	The aircraft must be loaded quickly including payload and battery.
M3: Launch Flight	$2 + N[\# \text{ laps flown} + \frac{\text{bonus box score}}{\text{test vehicle weight}}]$	Fuel tanks, flight battery, and X-1 test vehicle must be installed in 5 minutes. Aircraft must complete at least one lap and then launch X-1 test vehicle in designated area within the 5 minute flight window.	Aircraft must be capable of accurately deploying X-1 test vehicle. X-1 test vehicle should be able to land in designated area.
Ground	$N[\text{time}]$	Ground crew member must install fuel tanks and X-1 test vehicle. Aircraft then must successfully deploy X-1 test vehicle.	The loading of all attachments must be as fast as possible.

Table 3 Mission requirements.

3.2 Trade Studies: The team performed trade studies with the analytic hierarchy process to aid in the determination of the plane configuration. Tables 4 and 5 highlight some of the key structural determinations. Namely, the team decided on a monoplane aircraft with a high wing, conventional tail, and bow tricycle landing gear. The M2 fuel tanks will sit underneath the wings. The weights of each selection criterion are determined from the scoring and mission requirements located in Table 3. The team uses the decision matrices found in Tables 4 and 5 to determine the optimal aircraft configuration in conjunction with optimizer and sensitivity study results.

3.3 Preliminary Design and Sizing Results: WUDBF used the findings from the sensitivity analysis, optimizer, and trade studies to select an aircraft configuration. A high wing with a Hershey Bar shape was chosen for high stability, tendency to

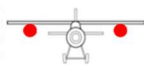
		Under Wing	Longitudinal	Stacked - Fuselage
				
Selection Criterion	Weight (%)	Rating (1-5)	Rating	Rating
Maneuverability	15	3	4	4
Stability and Control	30	5	3	3
Maximum Mass Capability	15	5	3	2
Attachment Efficiency	30	5	4	3
Flight Speed	10	2	4	4
Weighted Score		20.000	18.000	16.000
Rank		1	2	3

Table 4 Fuel tank payload decision matrix.

		High-Wing	Mid-Wing	Low-Wing
				
Selection Criterion	Weight (%)	Rating (1-5)	Rating	Rating
Maneuverability	15	3	4	5
Stability and Control	30	5	4	2
Takeoff length	20	3	4	5
Manufacturability	15	5	2	3
Flight speed	20	3	4	4
Weighted Score		3.900	3.700	3.600
Rank		1	2	3

Table 5 Wing placement decision matrix.

self-correct, and manufacturing simplicity. The wing will not include a dihedral, as one continuous carbon fiber spar will span the length of the wings, and a dihedral complicates manufacturing processes for minimal aerodynamic return. The spar will increase wing loading for M2. The wings have a span of 6 ft and a chord length of 1.457 feet. WUDBF selected a

conventional tail for stability and manufacturing efficiency. The horizontal stabilizer will have a span of 0.885 feet and a chord length of 0.787 feet. The vertical stabilizer will be tapered with a root chord of 0.689 feet and a tip chord of 0.328 feet.

A single propeller propulsion system was chosen because a takeoff limit will not be imposed. The team still aims to takeoff within 60 feet. This propulsion configuration was selected to achieve the highest efficiency and the lowest weight and drag. Additionally, this design allows for ease of avionics and propulsion work.

The fuselage will be a rounded rectangular prism with a flat top and bottom to allow for the installation of the wings and X-1 glider vehicle, "Ursa Minor". The fuselage will be 6 inches in height, 6 inches in width, and 28 inches in length to minimize weight, while allowing space for electronics and the possibility of an internal tank. The fuselage will be constructed with a molded fiberglass 4-ply layup and internal laser-cut plywood supports at the motor, front, and back for structural stability. All batteries will be inserted at the front of the aircraft through a hatch. Currently, WUDBF has decided against an internal fuel tank until construction and test flight of the first prototype. This is to verify design choices and determine if the aircraft can successfully carry more fuel or if there will be larger than acceptable adverse impact on aircraft performance. The team has chosen to use a Bow-Tricycle landing gear due to added structural strength of the rear of the fuselage and higher ground maneuverability. WUDBF plans to reinforce the front landing gear by attaching it to plywood reinforced with fiberglass. Figure 3 details the planned aircraft design.

The M2 fuel tanks will be carried underneath the wings. This was selected for the attachment efficiency and higher stability during flight. Lastly, the M3 X-1 Glider will be connected to the underside of the plane for efficiency and ease of deployment. The lights will use a circuit that is always powered. The connection will be interrupted when attached to "Ursa Major" to keep strobe lights off until deployment. A servo-lock will control the release of the glider. The position of the servo arm determines whether the glider is locked in place or deployed. WUDBF performed an XFLR stability test on glider designs and picked a set up with a one counter-rotational aileron, an elevator, and rudder. Each component will be controlled with one servo to minimize weight. The M3 glider will have a maximum footprint of 1.5 feet wide, 1.5 feet long, and 3 inches tall. This ensures that all the necessary components and electronics fit and function properly, while minimizing weight. Additionally, the height is kept small to minimize the drag added on the main aircraft by the glider.

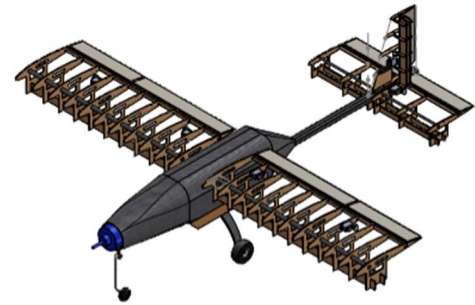


Figure 3 Proposed aircraft.

The propulsion pack will consist of a LiPo 8s (29.6-volt) 3250 milliamp hours battery, a SunnySky x3530 KV445 motor, and a 13x8 propellor. These choices are determined using eCalc setupFinder to select a battery and motor combination. eCalc propCalc is then used to analyze the performance of the system. The combinations with the best performance were selected for the aircraft.

3.4 Sensitivity Study: To determine which scoring variables have the biggest impact on competition performance, optimization and sensitivity analysis is performed using an in-house trade space optimizer written in MATLAB. The optimizer selects key variables determined to be most important from the mission requirements in Table 3. These variables include M2 fuel tank weight, M3 glider weight, wing area, and cruise velocity. After consulting the rules, WUDBF makes assumptions to set constraints. The optimizer then sweeps through different values for each of the key variables using the constraints. M2 and M3 sensitivity analysis results can be seen in Fig. 4. From sensitivity analysis, WUDBF

determines that it is important for scoring to maximize M2 fuel tank weight, maximize cruise velocity, and minimize M3 glider weight. The M3 assertion is confirmed by the optimizer's selection of the manually entered lower bound of the glider weight, determining the highest M3 score will be achieved by minimizing glider weight. The optimizer also determined that M2 fuel tank weight is more important than cruise velocity.

Based on the sensitivity analysis, it was determined that minimizing M3 glider weight should be the highest priority, as lowering glider weight would greatly increase competition score without harming the cruise velocity. Though maximizing M2 fuel tank weight and maximizing cruise velocity is important, it is not feasible to maximize one without decreasing the other. However, because M2 fuel tank weight is more important than cruise velocity, WUDBF prioritizes having a large wing area at the expense of increased drag.

Mission 1 (M1) only requires flying three laps successfully. Because there are no additional variables impacting M1 score, WUDBF has determined that M1 does not require any additional aircraft optimization. However, during optimization, the team still set a maximum take-off distance of 100 feet.

3.5 Design Approach: WUDBF has come up with a set of initial targets for contest scoring parameters based on sensitivity analysis and results yielding high scores from previous years. For Ground Mission (GM), at a minimum, the team aims to complete the cycle in under 5 minutes as this is required for the staging box at the flight line as well. Ideally, WUDBF hopes to cycle through configurations and complete GM in under 2 minutes.

M1 only requires flying three laps, so the team aims to successfully complete all three laps.

WUDBF determined that for M2, fuel tank weight is the most important variable, so the aircraft will have a larger wing area despite the lower cruise velocity. Once wing area was calculated, the team determined maximum fuel tank weight based on a self-imposed take off limit of 100 feet. WUDBF aims to carry a minimum of 5.51 pounds. Currently, the team plans to only use external fuel tanks, with two on each wing, for a total of four to distribute the weight.

The team determined that minimizing glider weight is the most important for M3. The glider will be constructed as light as possible and with a footprint just large enough to house any necessary electronics. WUDBF aims to land the glider in the 2.5-point bonus box and keep the glider weight under 0.25 pounds.

4 Manufacturing Plan

4.1 Preliminary Manufacturing Flow: The manufacturing flow is outlined in Fig. 5. The results from ground and flight testing will be used to inform redesigns and further optimization of additional prototypes.

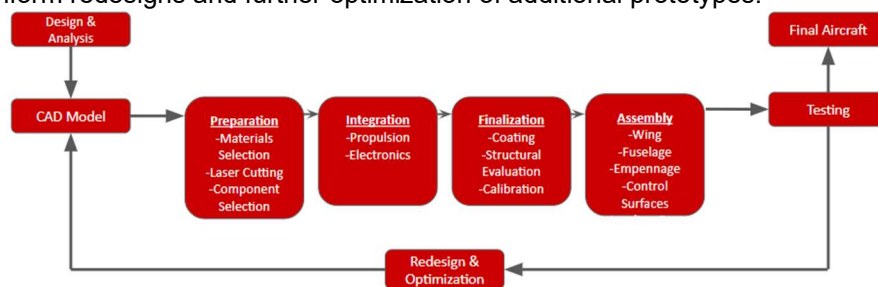


Figure 3 Manufacturing flow from prototype to final aircraft.

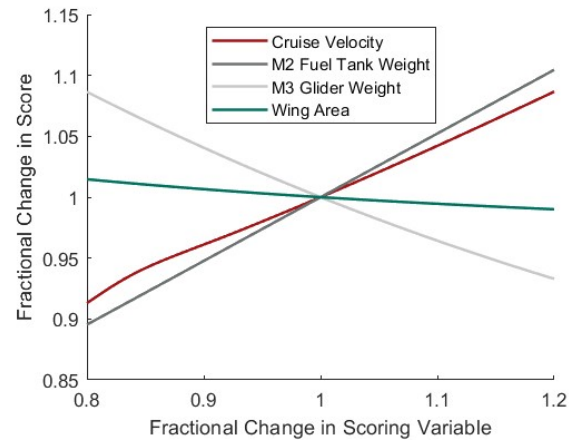


Figure 4 Sensitivity Study results for M2 and M3.

4.2 Critical Processes: Each section of the aircraft is broken down into individual components and modeled in SolidWorks. The wings and tail consist of a blend of balsa and birch ribs and spars. After completion of the wings and tail, they are given a balsa leading edge and covered in UltraCote exterior skin. Control surfaces are made up of XPS foam and covered in one layer of fiberglass for reinforcement. The fuselage is constructed from a 4-ply fiberglass wet layup done over top of a foam replica of the SolidWorks fuselage model, also of XPS foam. Fiberglass is selected due to its high strength and low weight in addition to its ability to control properties by varying the angles of each ply in the layup. The fuselage mold is broken up into 1-inch sections and cut out using a CNC hot-wire foam cutter. Each of these segments are glued together to form the full plug for the fiberglass. Additionally, WUDBF will use a commercially available carbon fiber rod for the tail boom due to high strength, stiffness, and low weight.

The X-1 glider will be constructed out of XPS foam due to ease of manufacturing, low weight, and the ability to increase strength and stiffness by covering with a layer of fiberglass. WUDBF plans to use three servos maximum to minimize weight. Additionally, strobe lights will cover the entire glider to ensure they are seen. The team will design a continuously powered circuit that is interrupted when attached to the main aircraft in-house.

5 Testing Plan

5.1-5.2 Component Test, Ground Test, and Flight Test Plan: Multiple tests will be conducted on individual aircraft components and test vehicles prior to manufacturing and testing of the final aircraft. These tests are shown below in Table 6. Various X-1 test vehicle configurations will be generated, constructed, and tested for additional optimization. WUDBF will simulate ground mission tests to ensure efficiency of aircraft set-up. The team has a thrust testing stand that allows testing of the "top" motor and battery combination. Additionally, pilot feedback from test flights in addition to data from the Pixhawk, pitot tube, and voltage sensor will be used to assess design and building process success.

Test Type		Objective	Method	Sub - Team
Preliminary Design Tests	Thrust Test	Choose the most efficient propeller, motor, and battery combination for the propulsion system. Compare with theoretical results from KMSim and ecalc.	Connect the top two motor choices to an RCBenchmark test stand and run thrust tests to select the best motor.	Propulsion
	Assembly Test	Minimize time spent assembling attachments for GM.	Practice ground mission with complete prototype.	Manufacturing
Ground Test	Wingtip Test	Verify the wing structure's resistance to failure under high loading stress.	Lift the plane carrying its Maximum Take Off Weight (MTOW) from the wingtips and check for damage.	Manufacturing
	CG Test	Verify that the CG is in the desired position.	Lift the plane from the wingtips at the calculated CG point for the desired static margin and check that the plane is level for all mission configurations.	Manufacturing
	Electronics Test	Ensure that all RC/X-1 test vehicle electronic systems work properly.	Connect all electronics as they will be in flight and test their functionality. Test the propulsion, control surfaces, and X-1 test vehicle lights.	Avionics
	Full Throttle Test	Verify that the motors are mounted securely when the throttle is at its max.	With a full prototype, hold the plane in place while pilot increases the throttle to the max. Check for any damage to the connection.	Manufacturing
	Landing Gear Test	Verify that the landing gear will not break when the plane lands.	Load the plane with its MTOW and perform a 2ft drop test. Check for any damage to the landing gear and any relevant connections.	Manufacturing
	X-1 Test Vehicle Deployment Test	Verify X-1 Test Vehicle has a reliable and accurately timed deployment mechanism.	Load the plane with the X-1 test vehicle and wire up the strobe lights. Perform drop tests from 2-3ft to ensure lights stay on and deployment is successful. Repeat to ensure consistency.	Avionics
	Stability Analysis	Verify the theoretical stability of the plane design, with and without fuel tanks/X-1 test vehicle.	Use XFLR5 to assess the aircraft's stability characteristics.	Aerodynamics
Flight Test	Take-off Distance Test	Ensure that the plane can take off within designated area (set by the team), with and without fuel tanks/X-1 test vehicle.	With the completed prototypes and final aircraft, a pixhawk (contains an accelerometer, gyroscope, barometer, and a magnetometer), pitot tube, a gps module, and a voltage and heat sensor will be used to quantitatively and qualitatively assess the plane's performance. This will confirm or reject design choices and inform any changes to the overall plane design or subcomponents. The final flight test will be used to estimate the competition score by simulating all four missions.	All Teams/Pilot
	Cruise Test	Assess the plane's handling and compare actual performance to expected results, with and without fuel tanks/X-1 test vehicle.		
	Turning Test	Assess the plane's turning capabilities and measure turn radii to compare to calculated results, with and without fuel tanks/X-1 test vehicle.		
	Ground Handling	Assess how the plane handles on the ground and upon landing, with and without fuel tanks/X-1 test vehicle.	Ensure properly functioning release mechanism, strobe light activation, and accuracy of landing. This will be done during test flights, both on the ground and in the air.	Avionics
	X-1 Test Vehicle Test	Assess X-1 test vehicle autonomous capabilities,		

Table 6 Ground, Component, and Flight test plan for 2024-2025.